SCALING ON ONE PAGE

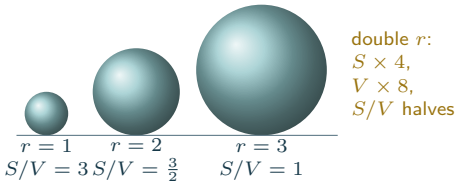
Surface to volume, and why small animals run hot

From the isoperimetric inequality to Kleiber's slope: a sphere gives $2/3$, and the observed $3/4$ asks for a fourth dimension.

1 / THE RULE AND THE PICTURE

The sphere has the least surface for its volume

The isoperimetric inequality $S^3 \geq 36\pi V^2$ becomes scale-free as a quotient, $\boxed{Q = \dfrac{36\pi V^2}{S^3} \leq 1}$ with $Q = 1$ for the sphere alone. Scaling every length by λ multiplies S by λ^2 and V by λ^3 and leaves Q unchanged, so Q records shape only, whereas for a sphere of radius r the ratio itself is $S/V = 3/r$.



double r: $S \times 4$, $V \times 8$, S/V halves

shape	$Q = \dfrac{36\pi V^2}{S^3}$	$S/V^{2/3}$
sphere	1	4.84
cylinder, $h = 2r$	$2/3$	5.54
cube	$\pi/6 \approx 0.52$	6.00
regular tetrahedron	$\pi/(6\sqrt{3}) \approx 0.30$	7.21
human, $70\,\mathrm{kg}$, $1.8\,\mathrm{m}^2$	≈ 0.095	≈ 10.6

2 / ANIMALS AS HOMOGENEOUS SPHERES

Heat leaves through the surface, so the slope is $2/3$

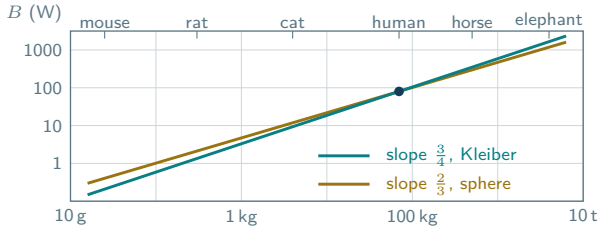
With uniform density ρ an animal of mass M has $r \propto M^{1/3}$. Heat escapes through the surface at a rate $\propto S \propto r^2 \propto M^{2/3}$, by Newtonian cooling $hS\,\Delta T$ or by radiation $\sigma\varepsilon S(T^4 - T_0^4)$, and a warm-blooded body holding its temperature burns at that same rate (Rubner's surface rule), so each kilogram burns at $B/M \propto S/V = 3/r$, giving $B \propto M^{2/3}$ and $B/M \propto 3/r \propto M^{-1/3}$, a straight line of slope $\frac{2}{3}$ in $\log B$ against $\log M$.

	M (kg)	r	S/V
mouse	0.025	$1.8\,\mathrm{cm}$	$165\ \mathrm{m}^{-1}$
human	70	$26\,\mathrm{cm}$	$11.7\ \mathrm{m}^{-1}$
elephant	4000	$98\,\mathrm{cm}$	$3.0\ \mathrm{m}^{-1}$

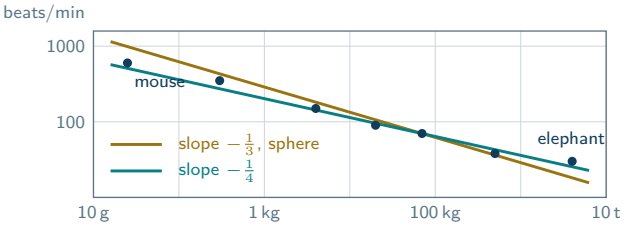
at $\rho = 1000\ \mathrm{kg\,m}^{-3}$; per kilogram the mouse loses heat $54 = (1.6 \times 10^5)^{1/3}$ times faster.

3 / KLEIBER'S OBSERVATION

Two straight lines on log–log axes, and the animals fall between them



Lines pass through a $70\,\mathrm{kg}$ human at $80\ \mathrm{W}$ and drift apart by 2.7 from mouse to elephant.



Approximate resting heart rates; the fitted slope -0.26 is nearer $-\frac{1}{4}$.

4 / COUNT DIMENSIONS

Which ball gives $3/4$?

A d-ball has $V = V_d r^d$ and $S = dV_d r^{d-1}$ ($V_3 = 4\pi/3$, $V_4 = \pi^2/2$), so $S/V = d/r$ in every dimension and, with $M \propto V$,

$$B \propto S \propto M^{(d-1)/d}, \qquad B/M \propto S/V \propto M^{-1/d}.$$

d	S/V	$B \propto$	$B/M \propto$
2	$2/r$	$M^{1/2}$	$M^{-1/2}$
3	$3/r$	$M^{2/3}$	$M^{-1/3}$
4	$4/r$	$M^{3/4}$	$M^{-1/4}$
5	$5/r$	$M^{4/5}$	$M^{-1/5}$

$$\frac{d-1}{d} = \frac{3}{4} \implies d = 4$$

5 / WHERE A FOURTH DIMENSION COULD COME FROM

The $1/r$ survives; only the mass law changes

Since $S/V = d/r$ keeps its $1/r$ form in every dimension, the difference between $2/3$ and $3/4$ lies in how mass relates to the length that sets the surface. West, Brown and Enquist (1997) found $B \propto M^{D/(D+1)}$ for a D-dimensional body fed by a space-filling, self-similar branching network, which gives $3/4$ at $D = 3$ and matches the ball formula once $d = D + 1$.

Measured exponents stay contested, with $3/4$ from Kleiber (1932) and values near $2/3$ from temperature-corrected mammal data (White and Seymour, 2003), so geometry shows which dimension $3/4$ needs and leaves open whether biology supplies it.

ASIDE / REAL ANIMALS ARE NOT SPHERES

Since $Q \leq 1$ the sphere is the best case for keeping warm. A human with $Q \approx 0.095$ exposes 2.2 times the sphere's area, which under geometric similarity shifts the prefactor of $B \propto S$ and leaves the exponent at $2/3$.

ASIDE / FIT THE SLOPE IN PYTHON

`np.polyfit(np.log10(M), np.log10(HR), 1)[0]` gives the slope of log rate against log mass, -0.265 for the seven points shown, since a power law $y = cx^p$ becomes the line $Y = pX + \log c$.

Convention. Heart rates and masses are approximate typical resting values, and both lines are anchored at a $70\,\mathrm{kg}$ human; Kleiber's fit is $B \approx 3.4\,M^{3/4}$ W for M in kg.
Companions: prime-box isoperimetric; logarithms lesson; scales essay.

Constructed by Silicon · Anthropic Claude — Architect Carbon · LeeMcCullochJames